

Daraz E-Commerce Usability Analysis

**A UX Evaluation and Statistical Analysis of
User Interaction and Product Search Efficiency**

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1 Introduction and Objectives

This study evaluated the usability of the Daraz.lk e-commerce platform using UX research methods and statistical analysis techniques. The project examined efficiency, effectiveness, memorability, and user satisfaction through moderated usability testing and hypothesis-driven analysis.

The purpose of this group usability study is to evaluate how effectively users interact with key features of [Daraz.lk](https://www.daraz.lk), a widely used e-commerce platform in Sri Lanka. Although Daraz provides a range of tools for product discovery and checkout, users often experience difficulties related to efficiency, effectiveness, memorability and trust. Understanding these usability aspects is essential for improving the overall shopping experience.

In this study, four members of the group investigate different usability metrics using different statistical tests. Each task focuses on a specific part of the user journey and examines how user performance or perception changes under varying conditions. The study aims to measure real user behaviour, identify usability problems and determine whether certain factors influence task outcomes.

The objectives of this usability evaluation are:

- **To measure efficiency** by comparing checkout completion time between experienced and inexperienced Daraz users (Independent-sample t-test).
- **To assess effectiveness** by examining whether users can find a product more successfully after applying filters (Paired-sample t-test).
- **To evaluate memorability** through a two-round product search and compare performance across conditions (ANOVA).
- **To examine satisfaction** by analysing the relationship between product ratings and user trust when viewing product pages (Pearson correlation).

Together, these objectives provide a structured understanding of how users perform different tasks on Daraz and how various factors such as experience level, filtering tools, repeated exposure and product rating affect their overall usability experience.

2 Research Questions and Hypotheses

2.1 Independent-sample-test

Student Name: Inuri Kolambage

Research question	Does checkout completion time differ between experienced and inexperienced Daraz users?	
Usability metric studied	Efficiency - measured by the time taken to complete the checkout process.	
Hypothesis (statement)	Null Hypothesis (H_0): There is no significant difference in checkout completion time between experienced and inexperienced Daraz users.	
	Alternative Hypothesis (H_1): There is a significant difference in checkout completion time between experienced and inexperienced Daraz users.	
Mathematical Expression	Null Hypothesis (H_0): $\mu_1 = \mu_2$ (μ_1 = Mean checkout completion time of experienced users)	
	Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (μ_2 = Mean checkout completion time of inexperienced users)	
Variables	Independent	User experience level - Experienced Daraz users - Inexperienced Daraz users
	Dependent	Checkout completion time - Measured in minutes and seconds from the task start to the task completion.
	Control	Same task instructions, Same moderator, The same timing method (stopwatch), Same meeting platform, Same checkout task
	Measurement Scale	Time taken to complete checkout - measured in minutes and seconds using a digital stopwatch.
	Data type	Ordinal, continuous data
Experimental design	Between-subjects design: Each participant belongs to only one experience group (experienced or inexperienced). This design avoids learning effects and ensures that the time difference is due to user experience, not memorability or familiarity gained from practice.	

Tasks and Scenarios	Participants were asked to complete a checkout process on Daraz by selecting any product they preferred and proceeding through the checkout steps until the final confirmation page (without making the payment). This task was designed to measure checkout completion time for experienced and inexperienced users.
Procedure	The usability test was conducted remotely through Google Meet, where each participant joined individually for the session. The test followed a moderated approach, with the same moderator guiding all participants to ensure consistency in instructions and timing. During each session, the moderator provided the task instructions, observed the participant's actions, and recorded the checkout completion time using a stopwatch. Moderation was necessary because the checkout process could involve confusion or hesitation, and having a moderator ensured clarity, accurate time measurement, and a consistent testing procedure across all participants.
Data collection method	Google Meet • Used to conduct all remote moderated usability sessions. • Allowed real-time observation of participant actions. Screen Recording (OBS) • Recorded each participant's screen during the checkout task. • Ensured accurate verification of task flow, timing, and user actions. Google Forms • Used to collect pre-test questions (experience level), in-test responses, and post-test feedback. • Ensured structured and organized data collection for analysis.

Test material	Participant Information Sheet (PIS) • Provided details about the study purpose, procedure, and participant rights. • Included the Informed Consent Form, which participants agreed to before starting the test. Software / Prototype • The official Daraz.lk website, used to perform the checkout task in a real shopping environment. Data Collection Instruments (DCI) • Task Scenario: Guided participants through the checkout process. • Screening / Pre-test Questions: Classified participants as experienced or inexperienced users. • In-test Questions: Captured task-related responses and noted difficulties during the process. • Post-test Questions: Collected feedback on user experience after task completion. • Screen Recordings: Captured participant actions for accurate timing and observation. • Timing Method: Checkout completion time recorded using the moderator's stopwatch or verified through recordings.
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2.2 Paired-sample-test

Student Name: Yelani Hanwella

Research question	Does applying filters improve user's ability to find the correct product on Daraz?	
Usability metric studied	Effectiveness	
Hypothesis (statement)	Null Hypothesis (H_0)	Applying filters does not improve user's effectiveness in finding the correct product on Daraz.
	Alternative Hypothesis (H_1)	Applying filters improves user's effectiveness in finding the correct product on Daraz.
Mathematical Expression	Null Hypothesis (H_0)	$\mu_1 = \mu_2$ μ_1 = Mean effectiveness score before applying filters (Task 1)

		μ_2 = Mean effectiveness score after applying filters (Task 2)
	Alternative Hypothesis (H_1)	$\mu_1 < \mu_2$ μ_1 = Mean effectiveness score before applying filters (Task 1) μ_2 = Mean effectiveness score after applying filters (Task 2)
Variables	Independent	Search condition: Whether the participant searches for the product without using any filters (Task 1) or using available filters such as price, rating, or brand (Task 2)
	Dependent	Effectiveness of product search (whether user found the correct product / task success / accuracy score)
	Control	Same product description, same website (Daraz), same task for both attempts, same participant performs both tasks

	Measurement Scale	Likert scales, 1=Very difficult, 5 = very easy
	Data type	Ordinal (Likert ratings and categorical ratings,suitable for paired-sample t-test when treated as interval data)
Experimental design	Within-subjects design: A within-subjects design was used because all participants were exposed to both conditions, searching for the product without filters and searching for the product using filters.	
Tasks and Scenarios	Start from the Home Page Task 1: Find the product without using any filters (search bar or categories only) Task 2: Find the same product again, this time using any filters (price, brand, rating, color, etc.) Scenario: User searches for a black digital unisex watch within Rs.1500–3000 and minimum 4-star rating.	
Procedure	Participants received instructions via Google Forms and completed the test remotely during a moderated Google Meet session. They first read the scenario, performed Task 1 (no filters), and answered	
	difficulty, confidence, and satisfaction questions. Then they returned to the Daraz home page, performed Task 2 (with filters), and answered the same question set again. Finally, they answered the post-test question about overall effectiveness. The test was moderated. The moderator provided clarification and assistance when participants needed help.	
Data collection method	Google Forms: The in-test and post-test surveys were administered through Google Forms to collect participant's difficulty ratings, confidence levels, satisfaction levels and overall effectiveness scores for both tasks. Google Forms provided a structured, uniform, and scalable method for gathering categorical and numerical data, while also enabling remote participation. Screen Recordings: Each usability session was recorded using a screen recording software while participants completed the tasks. After the session, the observer reviewed these recordings to confirm whether participants found the correct product, to assess how they navigated the Daraz website, and to observe how effectively they applied filters. This allowed accurate behavioural analysis without interrupting the participant's performance.	

Test material	Participant Information Sheet (PIS) : Participant Information Sheet, Informed consent form Software / Prototype: Daraz.lk website Data Collection Instruments (DCI):Task scenarios, Screening / Pre-test questions, In-test questions and data collected during the test, Post-test data collection
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2.3 ANOVA

Student Name: Bemash De Silva

Research question	Assessing user memorability in repeated product searches using a two-round item search task.	
Usability metric studied	memorability	
Hypothesis (statement)	Null Hypothesis (H_0): There is no significant difference in the time taken to complete the product search task between Round 1 (first attempt) and Round 2 (repeat attempt).	
	Alternative Hypothesis (H_1): The time taken to complete the product search task in Round 2 (repeat attempt) will be significantly less than in Round 1 (first attempt), demonstrating improved user memorability and efficiency.	
Mathematical Expression	Null Hypothesis (H_0): $\mu_1 = \mu_2$ μ_1 = Mean time taken to complete Round 1. μ_2 = Mean time taken to complete Round 2	
	Alternative Hypothesis (H_1): $\mu_1 > \mu_2$ μ_1 = Mean time taken to complete Round 1. μ_2 = Mean time taken to complete Round 2	
Variables	Independent	Round/Attempt Number: The participant's exposure to the task (Round 1 or Round 2).
	Dependent	Time taken to complete the task (measured in seconds). Also, Subjective Ratings (Difficulty, Confidence, Overall Easiness) collected via Likert scales.
	Control	Task Scenarios (The four specific search criteria are the same in both rounds), Website (Daraz), Participant (Same person performs all tasks/rounds), Break time, and Order of Tasks (Order of the four tasks should be consistent within a participant's two rounds, and ideally counterbalanced across participants).
	Measurement Scale	Ratio for time taken (time in seconds). Ordinal for subjective ratings (Likert scales 1 to 5).

	Data type	Continuous (Time in seconds). Ordinal (Likert ratings).
Experimental design	Within-subjects design: All participants are exposed to both the Round 1 and Round 2 conditions, and complete all four search tasks in both rounds. This is necessary to compare performance on the same set of tasks before and after a break to measure a learning/memorability effect.	
Tasks and Scenarios	Starting from the Daraz home page, the participant performs the full set of four unique search tasks sequentially in Round 1. After a short break, the participant repeats the exact same sequence of four tasks in Round 2. The four tasks are: 1. Find any item under Rs. 1500. 2. Find an item under Rs. 1500 with at least 4 stars. 3. Find an item under Rs. 1500 with free delivery. 4. Find an item under Rs. 1500 that has a discount.	
Procedure	Remote and Moderated. The moderator guides the participant through the sequence of 8 total tasks (4 in Round 1, 4 in Round 2). The moderator ensures the site is reset between tasks and rounds, confirms the participant is following the exact required search criteria for each task, and verifies the accurate start/stop of the timer for all 8 task times entered into the Google Form.	
Data collection method	Google Forms: Collects the 8 time recordings (4 per round) and the 8 sets of Subjective Ratings (4 per round), plus the Overall Easiness rating. Screen Recording/Observation: Used to verify success for all 8 tasks and to observe strategy changes across the two rounds.	
Test material	Data Collection Instruments : The Google Form must include distinct sections for the time and subjective ratings for each of the four tasks within both Round 1 and Round 2. Software/Prototype: Daraz.lk website	

2.4 Pearson Correlation

Student Name: Oshaka Dilhara

Research question	Is there a relationship between the product star rating displayed on a Daraz product page and the user's satisfaction and trust toward that product page? This research question evaluates whether higher product star ratings influence users' perceived reliability, accuracy, seller trust, and overall confidence when deciding whether a product can be trusted. These components collectively represent the usability metric of satisfaction.
Usability metric studied	Satisfaction is a subjective usability metric that reflects users' emotional and attitudinal responses when interacting with a system. In this study, satisfaction was measured through four aspects: ● Perceived reliability of information ● Accuracy of product details ● Trust in the seller ● Confidence in buying the product Each aspect was measured using 5-point Likert scales (1 = Strongly disagree → 5 = Strongly agree).
Hypothesis (statement)	Null Hypothesis (H_0): There is no linear relationship between product star rating and user satisfaction/trust.
	Alternative Hypothesis (H_1):
	There is a linear relationship between product star rating and user satisfaction/trust.
Mathematical Expression	Null Hypothesis (H_0): $H_0: \rho = 0$

	Alternative Hypothesis (H_1): $H_1: \rho \neq 0$	
Variables	Independent	Product Star Rating • 0.0 - 5.0 rating displayed on the Daraz product page • Measurement Scale: Ratio • Data Type: Quantitative, continuous
	Dependent	User Satisfaction / Trust Score Calculated by averaging the scores of: 1. Reliability 2. Accuracy of information 3. Trust in seller 4. Confidence in purchasing • Measurement Scale: Likert (ordinal, treated as interval for Pearson r) • Data Type: Quantitative
	Control	• All participants viewed the same three product pages (A, B, C) • Same instructions given to all participants • Same trust questionnaire used for each product • Same device/browser requirements
		• Same moderated environment (Google Meet)

	Measurement Scale	Independent Variable (Product Star Rating): Ratio scale ,star ratings are numerical, continuous, and have a true zero (0-5). Dependent Variable (User Trust Score): 5-point Likert scale (1 = Strongly disagree, 5 = Strongly agree). Although Likert items are ordinal, for Pearson correlation they are treated as interval data, as is standard practice in usability studies.
	Data type	• Star rating: continuous numerical (ratio) • Trust/Satisfaction: ordinal (converted to interval values for Pearson correlation)
Experimental design	A within-subjects correlational design was used. Every participant viewed three product pages Product A, Product B, Product C and rated each using the same four trust/satisfaction questions. Because each participant evaluated all three products: • Individual differences were controlled • Ratings could be compared consistently • The natural relationship between rating and trust could be measured This design is appropriate because Pearson correlation does not require manipulation only the measurement of a statistical relationship.	
Tasks and Scenarios	Task Given to Participants Participants completed the following steps for each product: 1. Open the product link provided in the Google Form. 2. Observe the star rating, reviews, seller information, and product details.	

	3. Rate their satisfaction/trust using four Likert items: ○ Reliability ○ Accuracy ○ Seller trust ○ Purchase confidence 4. Repeat the process for Product B. 5. Repeat the process for Product C. Scenario Provided to Participants “You will see several product pages. Please examine each product as you normally would before making a purchase. Look at the product rating, reviews, description, and seller details, then answer the trust/satisfaction questions honestly based on what you see.” This scenario ensured participants acted naturally, similar to real online shopping behaviour.
Procedure	The entire RQ4 test was conducted remotely using Google Meet and moderated to ensure all participants followed instructions. Exact Procedure 1. Moderator welcomed participants and explained the study. 2. Participants opened the Google Form and navigated to Section 6 (RQ4). 3. Participants shared their screens for accuracy. 4. For Product A: ○ Opened the link ○ Analysed the page ○ Answered the 4 satisfaction/trust questions 5. Repeated the same steps for Product B and Product C.

	6. Participants answered one final satisfaction question about Daraz product information accuracy. 7. Participants submitted the form while still sharing their screen to confirm data validity. Justification • Moderation prevents incorrect task execution • Screen sharing ensures authentic behaviour • Remote testing increases convenience and participation rate
Data collection method	Google Forms – for collecting trust/satisfaction ratings Google Meet – for real-time moderation and screen monitoring
Test material	For RQ4, the following materials were used: 1. Product Pages Three Daraz product pages with different star ratings: • Product A – High rating • Product B – Medium rating • Product C – Low rating 2. Trust/Satisfaction Questionnaire For each product, participants answered: 1. “I believe the information on this product page is reliable.” 2. “I think the product description and specifications are accurate.” 3. “I trust the seller to deliver the product as described.” 4. “I would feel confident buying this product based on this page.”

	(1 = Strongly disagree, 5 = Strongly agree) 3. Final Satisfaction Question “How satisfied are you with the accuracy of information on Daraz product pages?” 4. Appendices • Participant Information Sheet (Appendix I) • Moderator Instructions (Appendix II) • Links to product pages and questionnaire (Appendix IV) • Raw data and timestamps (Appendix VI)
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3 Study planning and organisation

3.1 Experimental Design

3.1.1 Experimental Design of Independent-sample-test (Q1)

Hypothesis: There is a significant difference in checkout completion time between experienced Daraz users and inexperienced Daraz users.

For this study, a Between-Subjects Design was selected to evaluate whether user experience level affects the efficiency of completing a checkout process in the Daraz website.

How the study is structured:

- Participants are divided into two groups based on their prior experience with the Daraz website:

Group 1: Experienced users, familiar with Daraz checkout processes.

Group 2: Inexperienced users, new to the website with no prior usage.

- Both groups perform the same checkout task on the Daraz website : selecting a random item and completing the checkout process using standard methods.
- The task completion time, measured from item selection to final checkout confirmation, is recorded for all participants.
- This design ensures clear, unbiased comparison of efficiency as measured by checkout time between the two user groups.

Task for Both Groups

All participants completed the same usability task:

- Search for a randomly assigned product
- Add the item to the cart
- Proceed through the checkout process using the standard Daraz flow (address selection, payment selection, order review)

This ensures consistency and comparability between the two groups.

Usability Metric

- Primary Metric: Task Completion Time

The time from the moment the participant begins searching for the assigned item until the moment they reach the final "Place Order" screen (without placing the order).

This metric aligns directly with the concept of efficiency, which is the usability metric under investigation.

Advantages of This Design

- Prevents contamination from familiarity or repeated interactions

- Ensures realistic and unbiased performance differences
- Allows clean statistical comparison using an independent-samples t-test
- Accurately reflects real variations between novice and expert users in real-world e-commerce environments

3.1.2 Experimental Design of Paired-sample-test (Q2)

Hypothesis: Users will be able to find the required product more effectively on Daraz after applying filters than when searching without using any filters.

For this hypothesis, a within-subjects experimental design was used. In experimental design, the investigator can fully control or manipulate the conditions so that a direct comparison can be made between two or more situations while other factors are ideally kept the same. In this study, the manipulated condition is the use of filters on Daraz.lk, and all other elements of the task remain constant (same product, same website, same participant).

Participants complete two tasks:

Task1 – No filters: Searching for the target product without using any filtering options.

Task 2 – With filters: Searching for the same product using any available filters.

Because the same participants experience both conditions, individual differences (such as experience, search behavior or familiarity with Daraz) do not affect the comparison. This makes the within-subjects approach the most suitable design for evaluating effectiveness and is consistent with the requirements of a paired-sample t-test, which analyzes differences in performance within the same group across two conditions.

Using this structure ensures that the only manipulated variable is the presence or absence of filters, allowing the study to effectively measure whether filters improve users' ability to locate the correct product.

Data Collection Method

Data was collected using a combination of questionnaire responses and observation tools. The Google Form was structured into pre-task, in-task, and post-task sections to capture paired measurements for both conditions

Pre-test questions:

- Scenario and task instructions
- Participant consent

In-test questions:

- Task 1 difficulty rating (5-point Likert scale)
- Task 2 difficulty rating (5-point Likert scale)
- Task completion confirmation
- Difficulties encountered

Post-test questions:

- Perceived effectiveness of filters
- Task preference (which task felt easier)
- Ease of learning filters

Observation tools:

- OBS Studio to record user interactions during both tasks

3.1.1.3. Experimental Design of ANOVA (Q3)

Hypothesis: The time taken to complete the set of product search tasks in Round 2 (repeat attempt) will be significantly less than in Round 1 (first attempt), demonstrating improved user memorability and efficiency.

A within-subjects experimental design was utilized for this study. This design ensures that every participant is exposed to both conditions (Round 1 and Round 2), allowing performance to be compared against the participant's own baseline.

The manipulated condition is the Round Number (Round 1 vs. Round 2), which directly tests the effect of memorability following repetition. All other variables, including the specific search criteria (the four tasks), the website (Daraz.lk), and the participant, are held constant.

The two task rounds are defined as:

- Round 1 – First Attempt: The participant performs the full sequence of four assigned search tasks for the initial time.
- Round 2 – Repeat Attempt: The participant repeats the exact same sequence of the four tasks after a short break, relying on memory of the successful steps from Round 1.

This within-subjects structure minimizes the influence of individual differences (such as varying search skills or website familiarity) on the results, isolating the effect of the independent variable (repetition) on the dependent variable (time taken). This structure is mathematically aligned with the requirements of a paired-sample t-test.

Data Collection Method

Data collection combined quantitative performance metrics (time) with qualitative feedback (ratings) using two primary methods.

1. Questionnaire Responses (Google Form)

The Google Form was structured to capture paired measurements for both Round 1 and Round 2 for each of the four tasks:

- Pre-test Section: Collected initial information, including the scenario, task instructions, and participant consent.
- In-test Questions (Collected for all four tasks in both Round 1 and Round 2):
 - o Time taken: Self-reported time (Ratio data) recorded by the participant for the completion of each task.
 - o Task Difficulty Rating: 5-point Likert scale rating (1 = Very difficult 5 =Very easy).
 - o Confidence Rating: 5-point Likert scale rating of confidence while performing the task.
- Post-test Questions (Collected once after Round 2):
 - o Overall Subjective Memorability: Single 5-point Likert scale rating on how much easier the second attempt felt compared to the first (1 =Much harder , 5 = Much easier).

2. Observation Tools

- Screen Recording Software (e.g., OBS Studio): Used to record the participant's screen and interactions during the performance of all tasks. The observational data served a crucial validation role to:
 - o Verify Task Success: Confirming that the participant successfully located a product meeting all criteria for all eight searches.
 - o Validate Time Data: Providing supplementary behavioral evidence to confirm the accuracy of the self-reported start and stop points for each task.

3.1.4 Experimental Design of Pearson Correlation (Q4)

A within-subjects correlational design was used because:

- Every participant viewed the *same three* product pages (A, B, C).
- Each participant provided trust ratings for each product.
- This design controls for individual differences (e.g., familiarity, experience).
- No variable is manipulated. we are measuring the natural association between rating and trust.

This structure is appropriate for Pearson's r , as it examines whether two numerical variables co-vary.

3.2 Procedure & Planning

3.2.1 Procedure and planning for independent-sample-test

Hypothesis: Experienced Daraz users complete the checkout process significantly faster than inexperienced users.

Test Duration:

Each individual testing session lasted approximately 15-20 minutes, which included:

- Welcome and brief introduction (2 minutes)
- Explanation of the study purpose and consent process (3 minutes)
- Pre-screening questionnaire to determine experience level (2 minutes)
- Main task execution and timing (5-10 minutes depending on participant performance)
- Post-task questionnaire completion (3-5 minutes)

The data collection involved testing 7 participants individually at times convenient to their schedules. Each individual testing session lasted approximately 15–20 minutes, allowing for a flexible yet consistent approach to accommodate participants' availability.

3.2.2 Procedure and planning for Paired-sample-test

Each test session followed a standardized sequence to ensure consistency:

1. Welcome and Consent (2–3 minutes)

- Participant was greeted and shown the Participant Information Sheet (see Appendix I).
- Informed consent was collected through the Google Form.
- The moderator briefly explained the study purpose and the two upcoming tasks.

2. Scenario Introduction (1 minute)

- Participants read the scenario describing the target product: A black unisex digital watch priced between Rs.1500–3000 with a minimum 4-star rating

3. Task 1 – Search Without Filters (3–5 minutes)

- Participants attempted to find the product without using any filters.
- Their interaction was recorded using OBS Studio for verification
- After completing Task 1, participants answered the in-task questions (difficulty rating, satisfaction, confidence).

4. Transition (1 minute)

- Participants were instructed to return to the Daraz home page to reset the interface.

5. Task 2 – Search Using Filters (3–5 minutes)

- Participants repeated the same search task, now allowed to use filters (price, rating, brand, etc.).
- OBS Studio continued to record the process.
- Participants then answered the second set of in-task questions.

6. Post-test Questionnaire (2–3 minutes)

- Participants responded to the post-task questions comparing both tasks and evaluating the usefulness of filters.

7. Session End (1 minute)

- Participants were thanked.

Test Duration

Each session took approximately 10–12 minutes, including consent, instructions, Task 1, Task 2 and the post-test questionnaire. However, the exact duration varied between participants depending on their individual performance, familiarity with Daraz, and the time they took to complete each task. Additionally, minor variations occurred due to occasional internet connection delays, which could influence how quickly pages loaded during the search tasks.

Why Participants Completed These Tasks

These tasks were specifically designed to evaluate the effectiveness hypothesis:

- Task 1 (no filters) measured baseline difficulty and effectiveness when searching manually.
- Task 2 (with filters) measured whether filters make the same search easier or more effective.

Because both tasks were completed by the same participants, the study produced paired data, enabling a direct before and after comparison using a paired-sample t-test.

Participant Information Sheet

All participants were provided with the PIS, which explained:

- The study purpose
- Procedures
- What data would be collected
- Voluntary participation and confidentiality
- Permission for screen recording

(See Appendix I. Participant Information Sheet)

Unmoderated Testing Considerations

If any sessions were unmoderated:

- The Google Form included clear step-by-step instructions for each task.
- Required question formatting prevented missing data.
- OBS recording verified participants followed instructions correctly.

3.2.3 Procedure and planning for ANOVA

Each test session followed a standardized sequence to ensure consistency for measuring the learning effect across the two rounds:

1. Welcome and Consent (2–3 minutes)

1. The participant was greeted, and the Participant Information Sheet was presented (see Appendix I).
2. Informed consent was formally collected via the Google Form.
3. The moderator briefly explained the study purpose: to understand user memorability when performing repeated search tasks on Daraz.

2. Scenario Introduction and Task Assignment (1–2 minutes)

1. The moderator introduced the overall task structure: the participant would complete a sequence of four different search tasks in Round 1 and then repeat the exact same sequence in Round 2.
2. The moderator confirmed the four search criteria that the participant would follow:
3. Find any item under Rs. 1500.
4. Find an item under Rs. 1500 with at least 4 stars.
5. Find an item under Rs. 1500 with free delivery.
6. Find an item under Rs. 1500 that has a discount.

3. Round 1 – Initial Attempt (6–8 minutes)

1. The moderator instructed the participant to begin the screen recording (via OBS Studio or similar).
2. The participant was instructed to start their own timer at the beginning of the first task.
3. Participants worked through the full sequence of all four search tasks consecutively. For each task, the participant was required to:
 - Find a product matching the criteria.
 - Stop the timer immediately upon finding the matching product.
 - Enter the time taken and answer the in-task questions (difficulty and confidence ratings) for that specific task in the Google Form before proceeding to the next one.
 - The site was reset to the Daraz home page before the start of each of the four tasks.

4. Short Break/Transition (1 minute)

- The participant was asked to take a very brief pause (e.g., 60 seconds) to simulate a short gap between uses, which helps assess true memorability rather than immediate short-term recall.
- The moderator confirmed the interface was fully reset to the Daraz home page.

5. Round 2 – Repeat Attempt (4–6 minutes)

- The participant was instructed to repeat the exact same sequence of four tasks in the same order as Round 1.
- For each of the four tasks, the participant followed the same timing and data entry procedure as in Round 1: start timer, complete task, stop timer, enter time and ratings.
- The interaction was recorded continuously via OBS Studio.

6. Post-test Questionnaire (2–3 minutes)

- The participant responded to the final post-task question evaluating the overall perceived ease of the second attempt compared to the first (the direct subjective measure of memorability).

7. Session End (1 minute)

- The participant was thanked for their time, and the screen recording was stopped.

Test Duration

Each session took approximately 14–18 minutes, encompassing consent, instructions, the eight individual search tasks (4 in Round 1 + 4 in Round 2), and the questionnaire responses. The overall duration varied depending on individual performance and task complexity, as the total time for four tasks in Round 2 was expected to be measurably shorter than in Round 1 due to the memorability effect.

Why Participants Completed These Tasks

These two rounds were specifically designed to evaluate the memorability hypothesis:

Round 1 (First Attempt): Established the baseline efficiency (time taken) and subjective difficulty for initially solving the complex search tasks.

Round 2 (Repeat Attempt): Measured the gain in efficiency and improvement in subjective ease, directly quantifying how well users remembered the successful search strategy and applied it.

Because both rounds were completed by the same participants, the study produced paired data (8 time measurements per participant), enabling a direct before-and-after comparison using a paired-sample t-test on the mean time taken across the four tasks.

Unmoderated Testing Considerations

As this was a moderated test (recommended for time-based data), the moderator ensured consistency. If any sessions were conducted unmoderated, the following measures were necessary:

The Google Form required clear, step-by-step instructions for performing and timing all eight tasks sequentially.

Required question formatting ensured no critical time or rating data was missing.

The OBS screen recording verified that participants followed the instructions (e.g., correctly resetting the site and following the specified order of the four tasks).

3.2.4 Procedure and planning for Pearson Correlation

For RQ4, the procedure was designed to measure whether a relationship exists between the product star rating displayed on Daraz and the user's satisfaction/trust when viewing those product pages. A standardised, moderated, and controlled process was followed to ensure that all participants completed the task consistently and provided valid data for Pearson correlation analysis.

1. Introduction and Briefing (1 minute)

Participants were informed that they would evaluate three different Daraz product pages. The moderator explained that the goal was to observe how the star rating influences satisfaction and trust. Participants proceeded to Section 6 of the Google Form while keeping their screen shared. No purchasing or interactions beyond viewing were required.

2. Accessing Product Pages (1–2 minutes)

Participants were provided with three product links labelled Product A, Product B, and Product C.

For each product page, participants:

- Opened the link
- Viewed the page while screen-sharing
- Observed the star rating, reviews, seller details, and description accuracy

This ensured all participants viewed the same controlled material under identical conditions.

3. Evaluation Task (4–5 minutes)

For each of the three products (A, B, and C), participants completed the same set of questions:

- Reported the product's star rating
 - This served as the independent variable (X).
- Rated satisfaction/trust using four 5-point Likert questions:
 - “The information on this product page is reliable.”
 - “The product description and specifications are accurate.”
 - “I trust the seller to deliver the product as described.”
 - “I would feel confident buying this product.”

The four responses were averaged to generate a trust/satisfaction score (Y).

These paired X-Y values are required for computing Pearson's correlation coefficient.

4. Post-Task Satisfaction Question (1 minute)

After evaluating all three products, participants answered one final question:

- “How satisfied are you with the accuracy of information on Daraz product pages?”

This provided overall user sentiment to support the interpretation of results.

5. Completion and Verification (1 minute)

Before submission, the moderator checked that all questions for Product A, B, and C were answered. Participants submitted the Google Form while screen sharing, ensuring data accuracy and completeness. Each submission was timestamped to maintain consistency and verification across sessions.

Test Duration

The RQ4 task took approximately 6–8 minutes per participant. This included:

- 1 minute for briefing
- 1–2 minutes for accessing the product pages
- 4–5 minutes for the evaluation and post-task questions

Slight variations occurred depending on internet speed and individual reading pace.

Why Participants Completed These Tasks

These tasks were specifically designed to generate the two variables required for the correlational hypothesis:

- Star rating (X) → objective system-generated value
- Trust/Satisfaction (Y) → subjective user rating

By collecting X-Y pairs for three different products per participant, the study was able to test whether higher star ratings on Daraz product pages are associated with higher user satisfaction and trust directly addressing the research question.

Participant Information Sheet

All participants received the Participant Information Sheet (see Appendix I) explaining:

- Study purpose
- Task requirements
- Use of collected data
- Voluntary participation
- Screen-recording permission

Consent was provided through the pre-test Google Form.

Unmoderated Testing Considerations

If any sessions were unmoderated:

- The Google Form included clear step-by-step instructions for each task.
- Required question formatting prevented missing data.
- OBS recording verified participants followed instructions correctly.

3.3 Roles & Responsibilities

Moderators: Yelani, Oshaka

- Guided participants through the entire usability session.
- Provided task instructions clearly and consistently for all participants.
- Ensured participants understood how to complete the Efficiency, Effectiveness, Memorability, and Satisfaction tasks.

- Maintained a neutral tone without influencing participant decisions.
- Ensured ethical guidelines were followed and informed consent was collected.

Facilitator: Inuri, Bemash

- Helped with setting up the environment (devices, links, stable internet).
- Recorded any unusual events (errors, disconnections, delays).
- Ensured the OBS screen recording, if used, was functioning properly.
- Provided backup technical support during sessions.rs: Inuri, Bemash

Observers: Yelani, Inuri

- Observed participant behaviour during the test.
- Noted hesitation points, navigation struggles, or misunderstandings.
- Recorded qualitative insights that support the interpretation of results.
- Ensured that no external bias or interference occurred during the task.
- Reported observations to the team for discussion before analysis.

Team Coordinators: Oshaka, Bemash

- Managed team communication and ensured all tasks and materials were ready on time.
- Coordinated scheduling of participants and tracked completed sessions.
- Ensured all data collected was organized in a shared structure before analysis.
- Verified that all sections of the study (pre-test, in-test, post-test) were functioning properly.

Planning for the Usability Session

Session Setup

- Sessions were conducted remotely, using participants' own devices.
- Google Forms were prepared for pre-test, in-test, and post-test data collection. • OBS Studio or built-in screen recording tools were prepared.

Technical Preparation

- Verified Daraz.lk links were functional before each session.
- Confirmed all survey questions, Likert scales, and instructions were working correctly.

- Ensured the tasks in the Google Form appeared in the correct order.

Participant Scheduling

- Participants were recruited and scheduled over multiple days.
- Session duration varied depending on participant performance and internet conditions.

Ensuring Consistency

- All moderators used the same script.
- Every participant completed the tasks in the same sequence.
- The same Google Form was used for all participants to standardize data collection. • Observers documented behaviours using the same observation template.

(For a thorough description of the roles and responsibilities given to moderators during the usability testing process, see Appendix II, Moderators' Information Sheet: Planning for the Usability Session)

3.4 Participant Profile and Recruitment

Participant Profile

A pre-test questionnaire was used to collect essential background information about participants before completing the usability tasks. This ensured that their experience levels, familiarity with online shopping, and comfort using Daraz.lk were clearly understood and could be connected to the effectiveness hypothesis.

The pre-test form included the following components:

1. Demographic Information

Participants were first asked to provide: Age group (18–24, 25–34, 35–44, 45–54, 55+)

This allowed the team to confirm that all participants were adults and to observe whether age might influence product-search behavior.

2. Online Shopping Experience

Participants reported their general online shopping experience level through the following questions:

- Have you used Daraz.lk or any online shopping website before?
 - Options ranged from “I use Daraz regularly” to “I have never used online shopping websites.”

- How often do you shop online? (Very often → Never)
- How confident are you when using online shopping websites? (1–5 scale)

These questions helped identify whether participants were:

- Experienced online shoppers
- Occasional shoppers
- Completely new to online shopping

3. Device Used for Shopping

Participants indicated which device they usually use for online shopping, selecting from:

- Mobile phone
- Laptop/computer
- Tablet

This helps explain potential differences in performance, as search speed and filter visibility vary across device types.

4. Previous Difficulties in Online Shopping

Participants were asked whether they had previously experienced difficulties when using Daraz or other shopping apps (Yes/No/Not sure)

If they said yes, they could specify difficulties such as:

- Searching for products
- Understanding product details
- Using filters
- Checking reviews
- Adding items to cart
- Checkout issues
- None

This helped the team predict which parts of the task might challenge different user profiles and how this could impact effectiveness.

Recruitment Methods

Participants were recruited through a combination of:

WhatsApp,

- Research team shared the Google Form link

- Participants received instructions
- Screen-recording guidance was sent
- Allowed quick communication and reminders

Email Invitations,

- Used for more formal participant communication
- Included the Participant Information Sheet (PIS) and consent details
- Ensured participants clearly understood the study requirements

Recruitment continued for several days until the required number of valid responses was collected. All participants had to confirm voluntary participation in the pre-test form before continuing.

(For recruitment evidence, refer to Appendix III. Participant Recruitment Evidence.)

3.5 Data Collection Methods

Study Setting

The usability study was conducted **remotely**, and all participants completed the test online using their own devices. Sessions were scheduled individually, and each participant joined a live moderated session where the research team guided them through the process.. **Moderation Approach**

The study was moderated, meaning the moderator actively guided participants by:

- giving task instructions clearly
- ensuring participants understood what to do
- reminding them to start/stop timers where necessary
- confirming they returned to the form after each task
- ensuring the screen was shared throughout the session
- resolving any difficulties during tasks

The moderator did not influence participant decision-making or search behaviour

Data Collection

- Quantitative Data
 1. Closed-ended questions in both pre- and post-surveys were used to gather quantitative data such as through ratings, the likert scale, and yes/no responses.
 2. Task completion metrics such as the time taken for each task were observed through the recordings.
- Qualitative Data

1. The screen recordings of the participants were used to later observe the user behavior.
 2. Open-ended survey questions such as suggestions and other similar questions were also used to gather quantitative data.
-
- Google Forms – main platform for delivering pre-test, in-test, and post-test questions and collecting all quantitative and qualitative responses.
 - OBS Studio – used by the moderator to observe participant behavior in real time (participants did not record screens).
 - Timer/Clock (Laptop or Phone) – used by participants only for tasks requiring time measurement (checkout task & memorability rounds).
 - Communication Platforms (WhatsApp / Email) – used to send instructions, schedule sessions, and share the testing link.

3.6 Research Instruments (materials) and Equipment

Tools and Materials Used

1. Daraz Website

- Purpose: Used as the real environment in which all tasks were performed, including product search, checkout, and reviewing product information.
- Justification: Testing on the actual website provides accurate, real-world results, ensuring participants interact with the interface they would normally use.

2. Stopwatch/Timer Application

- Purpose: Record task completion time for efficiency-related tasks.
- Justification: Task completion time is a key quantitative measure. Using a digital timer ensures precise and reliable measurement.

3. In-Test and Post-Test Questionnaires (Google Forms)

- Included: In-test questions captured task performance and efficiency, such as checkout completion time, perceived ease, confidence, satisfaction, and overall experience. Post-test questions collected additional qualitative feedback on difficulties, preferences, and overall impressions.
- Purpose: Gather both quantitative and qualitative data from participants.
- Justification: Google Forms ensured consistency across participants, provided structured data for analysis, and allowed participants to respond conveniently while capturing all information needed for testing the research questions.

4. Observation Recording

- Purpose: Notes were taken during the tasks to record participant hesitations, errors, or difficulties.
- Justification: Observation supported interpretation of the collected data and helped contextualize any challenges participants faced.

Equipment

1. Participant Devices

Participants used their own laptops or computers to access the Daraz website.

- Justification: Using participants' own devices ensures natural interaction and reflects typical online shopping behavior.

2. Stable Internet Connection

- Required for smooth functioning of the website during all tasks.

Why These Materials Were the Best Fit

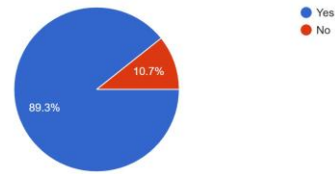
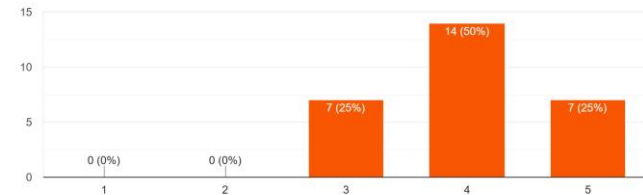
- The Daraz website provided a realistic context for performing tasks related to all research questions.
- The stopwatch ensured accurate timing for efficiency measures.
- Google Forms allowed consistent and structured collection of quantitative and qualitative data.
- Observation helped capture real-time behaviors and challenges.
- Using participants' own devices and a stable internet connection ensured natural behavior and minimal technical interruptions.

4 Study Results and Findings

4.1 Study results of Independent-sample-test

Research Question : Does checkout completion time differ between experienced and inexperienced Daraz users?

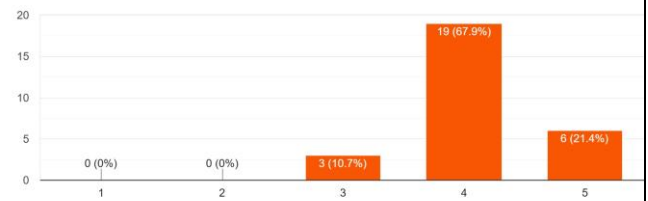
Usability Metric : Efficiency

In-test Questions																																																																																					
1. Have you ever made a purchase on Daraz before? This question identifies the participant’s level of experience with Daraz, which is the independent variable in our study. Participants were categorized as either experienced or inexperienced users based on their response. This classification allows for comparing efficiency and usability between the two groups.	1. Have you ever made a purchase on Daraz before? 28 responses  <table><tr><th>Response</th><th>Count</th><th>Percentage</th></tr><tr><td>Yes</td><td>25</td><td>89.3%</td></tr><tr><td>No</td><td>3</td><td>10.7%</td></tr></table>	Response	Count	Percentage	Yes	25	89.3%	No	3	10.7%																																																																											
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4. How confident were you that you completed the checkout correctly?

This question measures the participant's confidence in completing the checkout process accurately. Higher confidence among experienced users may indicate that familiarity with the platform helps them complete tasks with fewer mistakes and greater assurance.

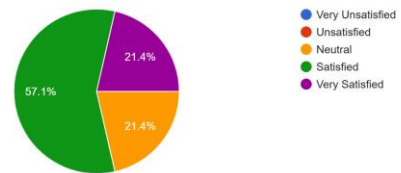
4. How confident were you that you completed the checkout correctly?
28 responses



5. How satisfied were you with the clarity of the checkout instructions?

This question measures the participant's satisfaction with how clear and understandable the checkout instructions were. Comparing responses between experienced and inexperienced users can help identify whether the instructions are communicated effectively to all users.

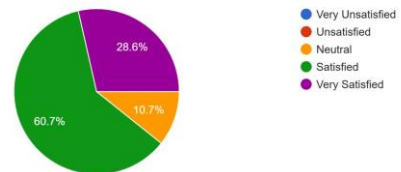
5. How satisfied were you with the clarity of the checkout instructions?
28 responses



6. Overall, how usable did you find Daraz's checkout process?

This question measures the participant's overall impression of the Daraz checkout process. Comparing responses between experienced and inexperienced users helps determine whether familiarity with the platform affects their general perception of the checkout experience.

6. Overall, how usable did you find Daraz's checkout process?
28 responses



4.2 Study results of paired-sample t-test

Research question : Does applying filters improve user's ability to find the correct product on Daraz?

Usability Metric: Effectiveness

In-test Questions

1. How difficult was it to find the product without filters?

What the data represents:

- This question captures the effectiveness score before applying filters (Condition 1).
- The rating reflects how difficult participants found the product-search task when no filters were allowed.

Relation to the Research Question:

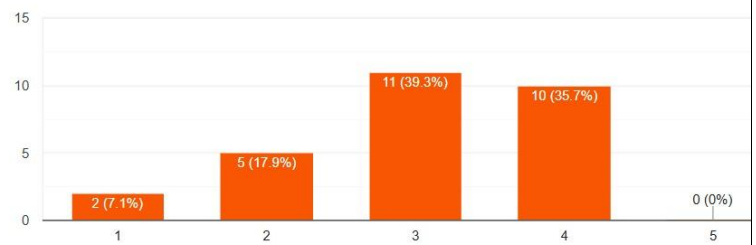
- Addresses the part of the research question evaluating effectiveness before filters.
- Provides the baseline needed to compare with the filtered condition.

Relation to Hypothesis and Variables:

- Dependent variable: Task difficulty (1–5 Likert scale)
- Independent variable: Search condition (No filters vs. With filters)
- This dataset forms Condition 1 in the paired-sample t-test to test whether filters improve effectiveness.

How this is used in analysis:

- These scores will be compared with the difficulty ratings after applying filters (Q2).
- Forms the first half of the paired dataset for the paired-sample t-test.



2. How confident were you that you found the correct product?

What the data represents:

- This question measures participants' confidence level during Task 1 (searching without filters).
- Confidence reflects how effectively participants felt they were able to identify the correct product under the no-filter condition.

Relation to the Research Question:

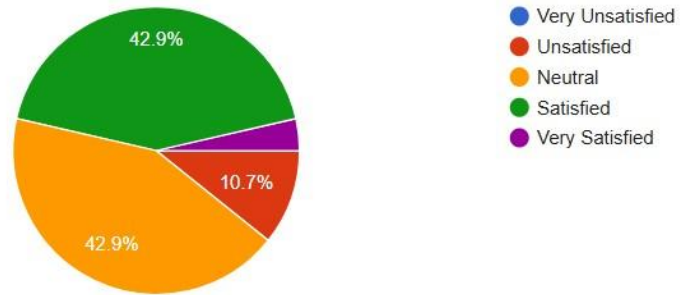
- Supports the research question by showing how effectively users can identify a product before applying filters.
- Helps determine whether lack of filters reduces user certainty or accuracy in selecting the correct item.

Relation to Hypothesis and Variables:

- Dependent variable: Confidence level (5-point Likert scale)
- Independent variable: Search condition (No filters vs. With filters)
- This dataset complements Q1 by capturing a second effectiveness indicator for Condition 1 before filters.

How this is used in analysis:

- Used to understand whether participants feel less confident in identifying products without filters.
- Together with post-filter confidence ratings, it helps evaluate whether confidence improves after filters supporting the paired-sample t-test analysis.



3. How satisfied were you with the search results you found?

What the data represents:

- This question captures participants' overall satisfaction after completing the product-search task without filters, indicating how well they felt the results matched what they were looking for.

Relation to the Research Question:

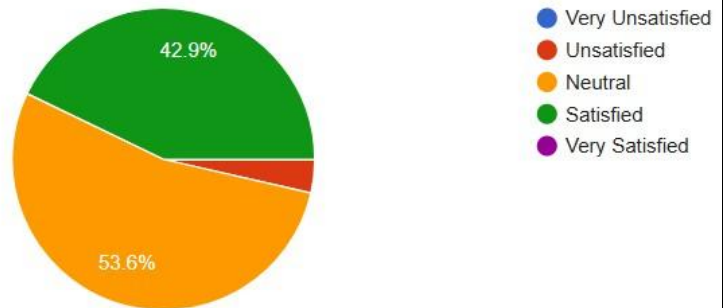
- This data provides the baseline satisfaction level for the "no-filters" condition. It is used to compare against the "after applying filters" condition to evaluate whether filters improve user effectiveness in finding the correct product.

Relation to Hypothesis and Variables:

- Independent Variable: Search condition (without filters vs. with filters)
- Dependent Variable: Satisfaction rating (Scale: 5-point Likert satisfaction scale. The satisfaction ratings from both tasks will be used in the paired-sample t-test to examine whether filters have a significant effect on perceived search effectiveness.)

How this is used in analysis:

- This satisfaction rating from Task 1 (without filters) is treated as an indicator of perceived effectiveness - how well users felt the search results matched what they intended to find. These baseline effectiveness scores will be paired with the effectiveness ratings from Task 2 (after applying filters).
- A paired-sample t-test will then compare the two conditions to determine whether using filters produces a statistically significant improvement in product-finding effectiveness.



4. How difficult was it to find the product with filters?

What the data represents:

- This question measures the perceived ease of finding the target product when filters are available.
- It reflects how effectively users could navigate and refine the product list using Daraz's filtering options

Relation to the Research Question:

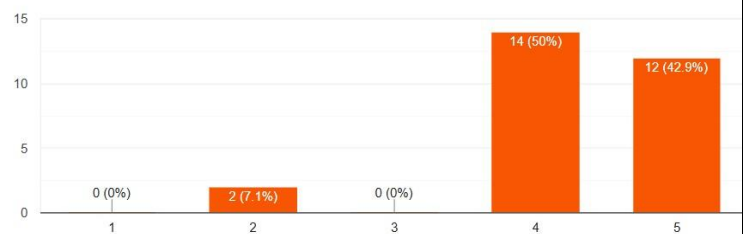
- This provides the effectiveness score for Task 2, which is the "with filters" condition. It directly supports the research question: "Do filters improve the effectiveness of finding a specific product on Daraz.lk?"
- Participants' ratings show a clear shift toward easier performance when filters are applied.

Relation to Hypothesis and Variables:

- Independent Variable:
Search condition (Task 1 = without filters, Task 2 = with filters).
- Dependent Variable:
Effectiveness, measured through participants' reported difficulty level when finding the product

How this is used in analysis:

- The difficulty ratings from Task 2 represent participants' perceived effectiveness after using filters.
- These values will be paired with Task 1 difficulty ratings (without filters) and analyzed using a paired-sample t-test.
- The test will determine whether the improvement in effectiveness (less difficulty to more effective search) is statistically significant



5. How confident were you that you found the correct product using filters?

What the data represents:

- This question measures how confident participants felt that they selected the correct product after applying filters.
- A majority reported high confidence (Satisfied or Very Satisfied), showing that filters helped users identify the correct product more easily and accurately.

Relation to the Research Question:

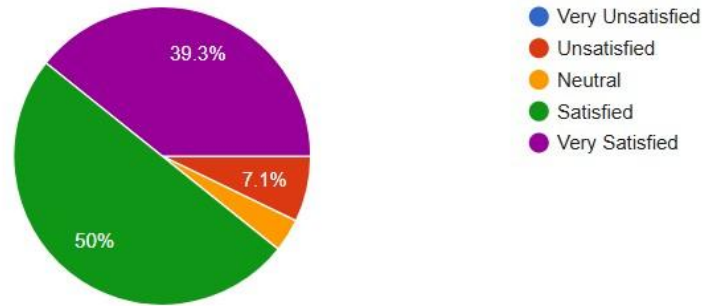
- The research question examines whether using filters on Daraz.lk improves user's product-finding effectiveness.
- Confidence in choosing the correct product is a direct indicator of effectiveness—higher confidence implies fewer mistakes and better search accuracy.
- Thus, this data directly answers whether filters make it easier for users to locate the right product.

Relation to Hypothesis and Variables:

- Independent Variable: Search condition (with filters)
- Dependent Variable: Effectiveness (measured through confidence rating)

How this is used in analysis:

- These Q5 confidence scores form the paired dataset (Condition 2) in the paired-sample t-test.
- They will be directly compared with Q2 confidence scores (Condition 1: without filters).
- If $Q5 > Q2$ significantly, it supports the hypothesis that filters improve product-finding effectiveness.



6. How satisfied were you with the filtered search results?

What the data represents:

- This question captures participants' satisfaction after completing Task 2 (finding the product using filters).
- The responses indicate how positively participants perceived the quality and usefulness of the search results produced after applying filters.

Relation to the Research Question:

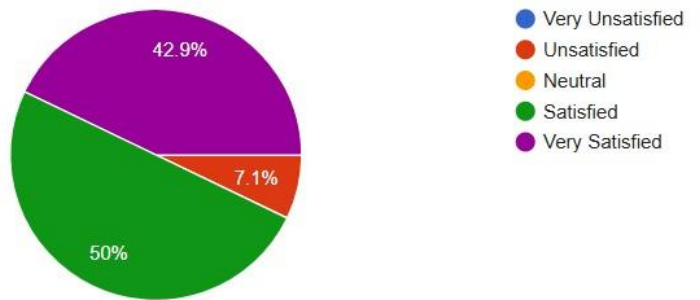
- This directly relates to the research question: "Does applying filters improve users' ability to find the correct product on Daraz more effectively?"
- Higher satisfaction suggests that the filtered method provides clearer, more accurate or more relevant results.

Relation to Hypothesis and Variables:

- Independent Variable: Search method (Before filters vs After filters)
- Dependent Variable: Satisfaction with search results (Likert scale 1–5). Satisfaction serves as a supporting effectiveness indicator, helping validate whether filtered search leads to a better user experience. Higher satisfaction after filters implies better effectiveness, complementing the primary effectiveness measure (task difficulty rating).

How this is used in analysis:

- Satisfaction scores for Task 2 help interpret whether improved effectiveness (lower difficulty) is accompanied by improved user perception.
- Although satisfaction itself is not part of the paired t-test, it provides qualitative support to interpret why the filtered task may yield better effectiveness outcomes.



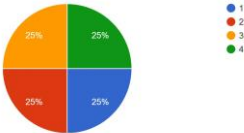
Higher satisfaction reinforces the expectation that Task 2 (with filters) performs better overall than Task 1 (without filters).																			
Post-test questions																			
7. Overall, which method felt more effective for finding the product? What the data represents: - Participants rated overall effectiveness on a 1–5 scale, where 1 = least effective and 5 = most effective. - Most participants rated 4 (28.6%) or 5 (60.7%), meaning the filtered method felt significantly more effective. Relation to the Research Question: - Addresses RQ2: “Does applying filters improve product-finding effectiveness on Daraz?” - Shows how participants perceived the improvement in effectiveness after using filters. Relation to Hypothesis and Variables: - Independent variable: Search method (without filters vs. with filters). - Dependent variable: Perceived effectiveness score (1–5). - Supports the paired-sample t-test hypothesis that Task 2 (with filters) would be rated as more effective than Task 1 (without filters). How this is used in analysis: - Serves as supporting subjective evidence that filters increase effectiveness. - While not included directly in the t-test (which uses the Task 1 & Task 2 difficulty scores), this question helps confirm the overall perceived improvement in effectiveness across participants. - Reinforces the direction of the	<table><thead><tr><th>Rating</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>0%</td></tr><tr><td>2</td><td>2</td><td>7.1%</td></tr><tr><td>3</td><td>1</td><td>3.6%</td></tr><tr><td>4</td><td>8</td><td>28.6%</td></tr><tr><td>5</td><td>17</td><td>60.7%</td></tr></tbody></table>	Rating	Count	Percentage	1	0	0%	2	2	7.1%	3	1	3.6%	4	8	28.6%	5	17	60.7%
Rating	Count	Percentage																	
1	0	0%																	
2	2	7.1%																	
3	1	3.6%																	
4	8	28.6%																	
5	17	60.7%																	

hypothesis by showing strong preference toward the filtered method.	
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4.3 Study results of ANOVA

Research Question : Assessing user memorability in repeated product searches using a two-round item search task.

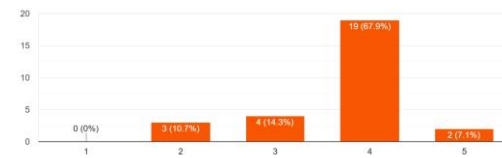
Usability metric : Memorability

In-test Questions	
Your category Participants were assigned to one of four task categories for the repeated item search tasks: - Find any item under Rs. 1500 - Find an item under Rs. 1500 with at least 4 stars - Find an item under Rs. 1500 with free delivery - Find an item under Rs. 1500 that has a discount Each participant was assigned one of these categories for all four tasks, ensuring that they performed the same type of task across both Round 1 and Round 2. This assignment allowed the study to examine memorability within specific task conditions, as task complexity and constraints may influence how easily participants remember task steps	Your category 28 responses 
1.Time taken to complete Round 1 This measures the duration participants took to complete the search tasks during the first round. It serves as a baseline for comparing performance in	1.Time taken to complete Round 1 28 responses 
Round 2 and is a key dependent variable in analyzing memorability.	

2. How difficult was the task?

Participants rated the perceived difficulty of each task during the first round. This helps identify initial challenges and establishes a baseline for comparing perceived difficulty in Round 2.

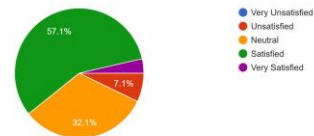
2. How difficult was the task?
28 responses



3. How confident were you while performing the task?

This measures participants' confidence in completing the tasks correctly during the first round. It provides insight into task familiarity and perceived mastery before repeated exposure.

3. How confident were you while performing the task?
28 responses



Round 2

4. Time taken to complete Round 2

This measures how long participants took to complete the same set of search tasks during the second round. Comparing this with Round 1 helps determine whether participants performed the tasks more quickly after having done them once before. Shorter times in Round 2 may indicate improved memorability of the task steps.

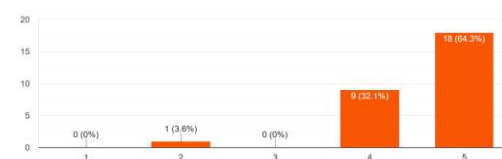
4. Time taken to complete Round 2
28 responses



5. How difficult was the task?

Comparing perceived difficulty between rounds allows us to evaluate whether participants found the tasks easier after prior exposure, which directly indicates memorability.

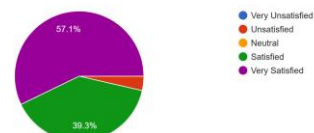
5. How difficult was the task?
28 responses



6. How confident were you while performing the task?

Confidence ratings in Round 2 show whether participants feel more assured in completing the same tasks, which is another indicator of memorability.

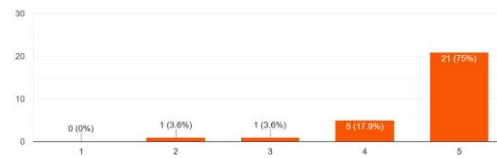
6. How confident were you while performing the task?
28 responses



7. Overall, how much easier did the second attempt feel compared to the first?

This question captures participants' subjective assessment of how much the tasks improved in ease after the first round. High scores indicate stronger memorability and effective retention of task steps.

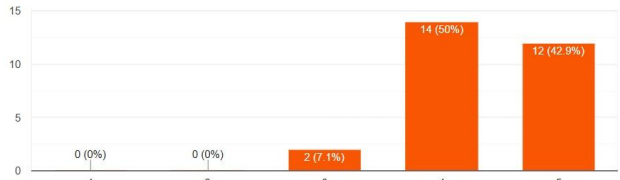
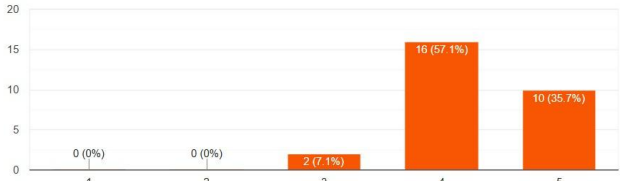
7. Overall, how much easier did the second attempt feel compared to the first?
28 responses



4.4 Study results of Pearson Correlation

Research Question : Is there a relationship between the product star rating displayed on a Daraz product page and the user's satisfaction and trust toward that product page? **Usability metric :** Satisfaction

In-test Questions																			
1.Product A - What is the displayed star rating? Result: All participants selected 5 stars for Product A (100%). This matches the actual rating shown on the Daraz product page. What the data represents: This is the independent variable (X) for Product A. A high star rating provides the baseline for testing whether trust/satisfaction scores also remain high. Relation to the Research Question: If higher star ratings lead to higher trust, then Product A should yield the highest trust values.	1. Product A – What is the star rating displayed on the product page? (ex. 4.9) 28 responses <table><tr><th>Star Rating</th><th>Count</th><th>Percentage</th></tr><tr><td>5</td><td>21</td><td>75%</td></tr><tr><td>5.0</td><td>7</td><td>25%</td></tr></table>	Star Rating	Count	Percentage	5	21	75%	5.0	7	25%									
Star Rating	Count	Percentage																	
5	21	75%																	
5.0	7	25%																	
2. Product A - I believe the information on this product page is reliable. Result: Most participants selected 4 or 5, indicating strong perceived reliability. What the data represents:	2. Product A – I believe the information on this product page is reliable. 28 responses <table><tr><th>Reliability Rating</th><th>Count</th><th>Percentage</th></tr><tr><td>1</td><td>0</td><td>0%</td></tr><tr><td>2</td><td>0</td><td>0%</td></tr><tr><td>3</td><td>0</td><td>0%</td></tr><tr><td>4</td><td>14</td><td>50%</td></tr><tr><td>5</td><td>14</td><td>50%</td></tr></table>	Reliability Rating	Count	Percentage	1	0	0%	2	0	0%	3	0	0%	4	14	50%	5	14	50%
Reliability Rating	Count	Percentage																	
1	0	0%																	
2	0	0%																	
3	0	0%																	
4	14	50%																	
5	14	50%																	

This item contributes to the trust/satisfaction score (Y). It measures perceived reliability of Product A. Relation to the Hypothesis: High reliability scores support the expectation that high rating → high trust.																			
3. Product A - I think the product description and specifications are accurate. Result: Responses were mostly 4s and 5s, showing users believed the information was accurate. What the data represents: This item adds to the accuracy component of the overall trust score. Relation to the Research Question: Indicates that users associate high star ratings with more accurate product information.	3. Product A – I think the product description and specifications are accurate. Copy chart 28 responses  <table><thead><tr><th>Rating</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>0%</td></tr><tr><td>2</td><td>0</td><td>0%</td></tr><tr><td>3</td><td>2</td><td>7.1%</td></tr><tr><td>4</td><td>14</td><td>50%</td></tr><tr><td>5</td><td>12</td><td>42.9%</td></tr></tbody></table>	Rating	Count	Percentage	1	0	0%	2	0	0%	3	2	7.1%	4	14	50%	5	12	42.9%
Rating	Count	Percentage																	
1	0	0%																	
2	0	0%																	
3	2	7.1%																	
4	14	50%																	
5	12	42.9%																	
4. Product A - I trust the seller to deliver the product as described. Result: Ratings were mostly Satisfied (4) and Very Satisfied (5). What the data represents: Measures the seller-trust element of the trust score. Relation to Hypothesis: Supports a positive relationship between star rating and seller trust.	4. Product A – I trust the seller to deliver the product as described. Copy chart 28 responses  <table><thead><tr><th>Rating</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>0%</td></tr><tr><td>2</td><td>0</td><td>0%</td></tr><tr><td>3</td><td>2</td><td>7.1%</td></tr><tr><td>4</td><td>16</td><td>57.1%</td></tr><tr><td>5</td><td>10</td><td>35.7%</td></tr></tbody></table>	Rating	Count	Percentage	1	0	0%	2	0	0%	3	2	7.1%	4	16	57.1%	5	10	35.7%
Rating	Count	Percentage																	
1	0	0%																	
2	0	0%																	
3	2	7.1%																	
4	16	57.1%																	
5	10	35.7%																	

5. Product A - I would feel confident buying this product based on this page.

Result:

Responses were mainly 4 and 5, indicating high purchase confidence.

What the data represents:

Final component of the trust score for Product A.

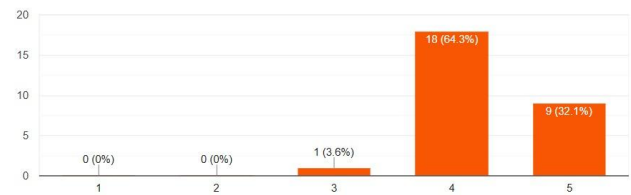
Relation to Hypothesis:

Again supports the expected positive relationship.

5. Product A – I would feel confident buying this product based on this page.

[Copy chart](#)

28 responses



6. Product B - What is the star rating displayed on the product page?

Result:

All participants selected 4.4 stars, reflecting a medium rating.

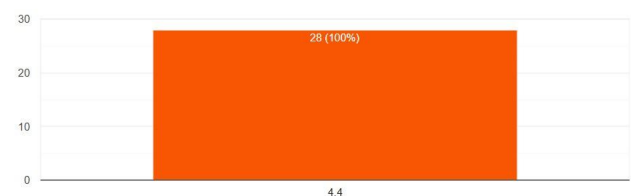
What the data represents:

This is the independent variable (X) for Product B.

6. Product B – What is the star rating displayed on the product page?
(ex. 4.9)

[Copy chart](#)

28 responses



7. Product B - I believe the information on this product page is reliable.

Result:

60.7% selected 4

32.1% selected 3

7.1% selected 2

Explanation:

Reliability scores are moderate, matching Product B's medium star rating.

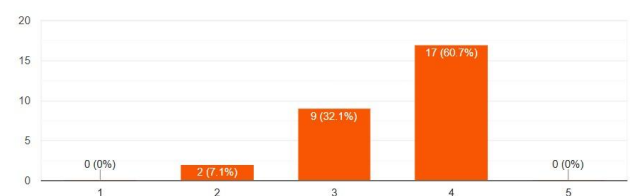
Relation to Hypothesis:

When the rating decreases ($A \rightarrow B$), reliability also decreases, supporting a positive correlation.

7. Product B – I believe the information on this product page is reliable.

[Copy chart](#)

28 responses



8. Product B - I think the product description and specifications are accurate.

Result:

2 → 7.1%

3 → 35.7%

4 → 46.4%

5 → 10.7%

Explanation:

Accuracy scores cluster around 3–4, showing acceptable but not excellent perceived accuracy.

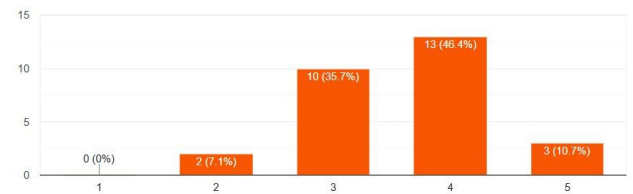
Relation to Hypothesis:

Medium rating → medium accuracy scores, matching the expected pattern.

8. Product B – I think the product description and specifications are accurate.

[Copy chart](#)

28 responses



9. Product B - I trust the seller to deliver the product as described.

Result:

2 → 10.7%

3 → 39.3%

4 → 50%

Explanation:

Users consider the seller moderately trustworthy lower than Product A but still positive.

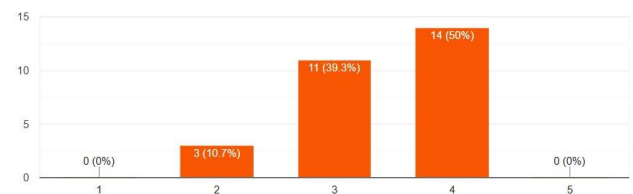
Relation to Hypothesis:

Trust decreases as rating decreases, aligning with a positive linear relationship.

9. Product B – I trust the seller to deliver the product as described.

[Copy chart](#)

28 responses



10. Product B - I would feel confident buying this product based on this page.

Result:

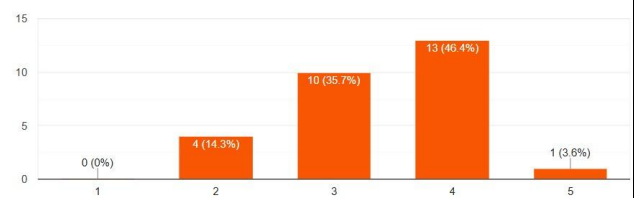
Most responses were 3 and 4, showing moderate purchase confidence.

Relation to Hypothesis:

10. Product B – I would feel confident buying this product based on this page.

[Copy chart](#)

28 responses



Confidence reflects the mid-range star rating.

11. Product C - What is the star rating displayed on the product page?

Result:

All participants selected 3 stars for Product C, with 100% choosing 3.

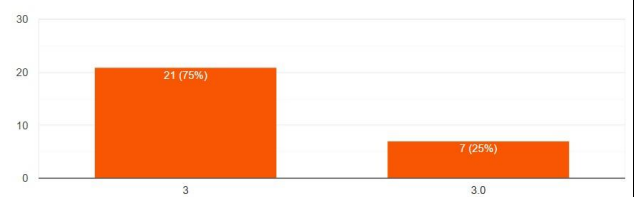
Explanation:

This confirms Product C as the low-rating product in the dataset.

11. Product C – What is the star rating displayed on the product page?
(ex. 4.9)

[Copy chart](#)

28 responses



12. Product C - I believe the information on this

Result:

Most participants gave low reliability ratings for Product C.

44.4% selected 2

29.6% selected 3

25.9% selected 1

No one selected 4 or 5.

Explanation:

Participants generally felt that the information on Product C's page was not reliable.

The responses show low confidence in the product details and overall trustworthiness.

Relation to the Hypothesis:

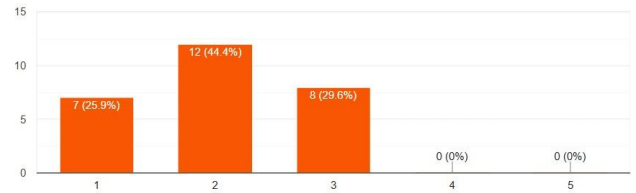
Product C had the lowest star rating, and it also received the lowest trust scores.

This supports the hypothesis that lower product ratings are associated with lower user trust.

12. Product C – I believe the information on this product page is reliable.

[Copy chart](#)

27 responses



13. Product C - I think the product description and specifications are accurate.

Result:

Most participants gave low accuracy scores for Product C.

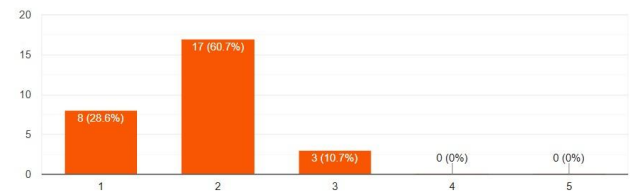
1 star: 8 responses (28.6%)

2 stars: 17 responses (60.7%)

13. Product C – I think the product description and specifications are accurate.

[Copy chart](#)

28 responses



3 stars: 3 responses (10.7%) 4 stars: 0% 5 stars: 0% Explanation: Participants generally felt that Product C’s description and specifications were not very accurate, with the majority selecting 1 or 2. Since Product C had a low star rating (3.0), users clearly associated this lower rating with lower confidence in the accuracy of its product details. Relation to the Pearson Correlation (RQ4): These low trust/accuracy ratings (Y variable) for a low-rated product (X = 3 stars) support the expected positive correlation between product rating and user trust: Lower star rating → lower accuracy and trust scores.																			
14. Product C — I trust the seller to deliver the product as described. Result: Most participants reported very low trust in the seller for Product C. 39.3% rated it 1 (Strongly disagree) 53.6% rated it 2 (Disagree) Only 7.1% rated it 3, and no participants selected 4 or 5. Explanation: This indicates that users did not trust the seller at all for Product C. Because Product C had a low star rating (mostly 3-stars), participants showed low reliability, low accuracy perception, and low seller trust. This supports the Pearson correlation expectation: Lower star ratings → Lower trust scores This aligns with the hypothesis that product rating (X) and user trust (Y) tend to move together.	14. Product C — I trust the seller to deliver the product as described. 28 responses <table><thead><tr><th>Trust Level</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>1</td><td>11</td><td>39.3%</td></tr><tr><td>2</td><td>15</td><td>53.6%</td></tr><tr><td>3</td><td>2</td><td>7.1%</td></tr><tr><td>4</td><td>0</td><td>0%</td></tr><tr><td>5</td><td>0</td><td>0%</td></tr></tbody></table>	Trust Level	Count	Percentage	1	11	39.3%	2	15	53.6%	3	2	7.1%	4	0	0%	5	0	0%
Trust Level	Count	Percentage																	
1	11	39.3%																	
2	15	53.6%																	
3	2	7.1%																	
4	0	0%																	
5	0	0%																	

15. Product C - I would feel confident buying this product based on this page.

Result:

Most participants selected 2 (50%) and 1 (35.7%), with only a small number selecting 3 (14.3%), and no responses for 4 or 5.

Explanation:

Participants reported very low confidence in buying Product C.

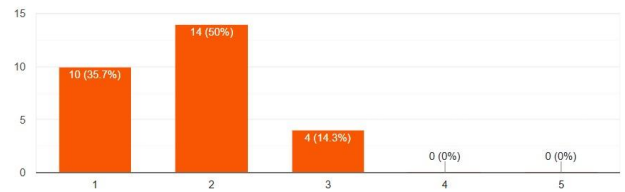
Since Product C had a low star rating (e.g., 3 stars), users did not feel confident about purchasing it.

This demonstrates that as the product rating decreases, user confidence also decreases showing a negative association consistent with the Pearson correlation expectation.

15. Product C – I would feel confident buying this product based on this page.

[Copy chart](#)

28 responses



16 .How satisfied are you with the accuracy of information on Daraz product pages?

Result:

Most participants selected Satisfied (46.4%) and Neutral (42.9%), while only a small number selected Very Satisfied (7.1%) and Unsatisfied (3.6%).

No participants selected Very Unsatisfied.

Explanation:

Overall satisfaction with product-page accuracy is moderately high, but not extremely high. Most participants trust the accuracy of Daraz product information to a reasonable extent, but still show some hesitation.

This supports the correlation finding because:

Products with higher star ratings (Product A & B) received higher trust and satisfaction scores.

The overall satisfaction pattern reinforces the idea that product rating influences user satisfaction and trust.

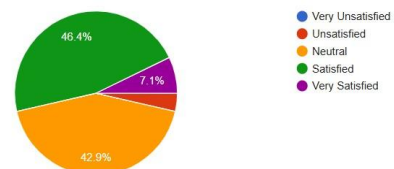
Thus, this final question aligns with the hypothesis:

Higher product ratings are generally associated

16 .How satisfied are you with the accuracy of information on Daraz product pages?

[Copy chart](#)

28 responses



with higher user trust and satisfaction.	
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5 Ethics form

We ensured ethical compliance by implementing the following:

- **Data Protection:** The collected data from surveys and screen recordings was securely stored, accessible only to the research team and used only for research purposes, where findings were presented in aggregate form.

These measures were taken to ensure that participants provided information voluntarily, under their privacy, and were not treated in a less than secure way regarding their data.

6 Conclusions

This usability study provided a structured understanding of how users interact with Daraz.lk when performing common online-shopping tasks. By combining pre-test profiling, moderated task performance and post-test reflections, the study successfully gathered reliable data to evaluate the overall usability of the platform. The testing process demonstrated that participants were generally able to navigate the site, complete tasks and provide meaningful feedback about their experience. Using multiple data-collection methods screen observation, structured questionnaires, and user feedback helped ensure that the findings were both comprehensive and consistent across all four usability dimensions.

Reflecting on the process, the moderated testing setup proved useful for maintaining consistency and clarifying instructions when needed. However, several improvements could have strengthened the study further. Additionally, conducting pilot sessions might have helped refine the wording of some questions and reduce participant confusion during transitions between tasks. More controlled test environments could also have reduced variations caused by individual device performance or internet connectivity.

Certain limitations were unavoidable in the experimental setup. Although moderation helped reduce misunderstandings, it may also have introduced observer effects, where participants behave differently because they know they are being monitored. The remote nature of the test meant that participant environments varied, influencing their comfort level, distractions and familiarity with the device they used. Another limitation was the

reliance on self-reported difficulty and satisfaction scales, which are subjective and may be influenced by personal bias, mood or social desirability.

Despite these constraints, steps were taken to minimize potential biases. All participants received the same scripted instructions, the tasks were presented in a fixed order to avoid confusion and the same digital tools were used for data collection. Screens were observed only for task verification and all participants answered identical pre-test, in-test and post-test questions to ensure consistency in measurement.

Overall, the study successfully captured meaningful insights into how users navigate Daraz.lk, how they perceive tasks difficult, and how filtering, navigation and product information affect their experience. While there are areas where the methodology could be further strengthened, the findings provide a valuable foundation for improving the usability of the platform and guiding future user-centred enhancements.